Machine learning to predict tennis match outcomes

Introduction:

Professional tennis match outcomes are influenced by a combination of measurable factors, including player rankings, recent performance, surface specialization, and matchup dynamics. At the same time, match outcomes remain inherently uncertain, as even heavily favored players can lose. This creates a natural question: to what extent can pre-match information be used to accurately predict match outcomes?

Player rankings alone provide a strong baseline for prediction, as higher-ranked players win a majority of matches on the ATP Tour. However, rankings are an imperfect measure of player ability and do not fully capture factors such as recent form, surface-specific performance, or serve style matchups.

This paper develops and evaluates machine learning models to predict ATP tennis match outcomes using only information available prior to each match. The analysis compares a simple ranking-based benchmark with a penalized logistic regression model (LASSO) and a nonlinear gradient boosted tree model (XGBoost). The objective is to assess whether more flexible modeling approaches can meaningfully improve prediction beyond rankings alone and to better understand the sources of predictive power in tennis match outcomes.

The results show that rankings explain a substantial portion of variation in match outcomes, and that machine learning models provide only modest improvements in overall predictive accuracy. However, the gradient boosted tree model produces better probability estimates and improves prediction in more competitive matches, suggesting that nonlinear relationships play an important role in determining match outcomes.

(This may be due to me selecting the model based on AUC and not purely accuracy. Models were tuned on multiple metrics yet I selected based on AUC.)

Data:

The final modeling dataset was built by combining multiple sources:

1. ATP Match Data

Historical ATP tour-level match data from Jeff Sackmann's publicly available tennis datasets. These files include:

- Match winners and losers
- Tournament name, level, surface, round, and date
- Match statistics (aces, serve percentages, etc.)
- Rankings and ranking points

2. Tournament Location Data

Tournament locations were manually standardized and geocoded in order to merge geographic and weather information.

3. NASA POWER Weather Data

Daily historical weather observations were collected using tournament coordinates and match dates, including:

- Temperature
- Humidity
- Wind speed
- Precipitation

4. Player Ranking Histories

Weekly ATP rankings and points were processed to create lagged ranking features available before each match.

The final data set was cleaned and constructed as follows:

The project required extensive data preparation:

- Combined yearly ATP match files into a one dataset
- Restricted analysis to the modern tennis era
- Standardized tournament names and locations
- Geocoded tournament cities and countries
- Merged weather data by location and date
- Constructed lagged ranking and points variables

- Removed post-match information to prevent data leakage
- Created a final modeling dataset using only pre-match predictors

Feature Engineering

A large set of pre-match predictors was engineered, including:

Player Quality Measures

- ATP ranking difference
- Ranking points difference
- Elo rating difference
- Surface-specific Elo difference

Recent performance Measures

- Wins in recent matches
- Rolling win percentages
- Ranking momentum and recent changes

Matchup Measures

- Head-to-head record
- Prior meetings count
- Serve versus return matchup variables

Workload / Fatigue Measures

- Matches played recently
- Surface experience
- Recent activity volume

Tournament and Context Variables

- Surface (hard, clay, grass)
- Tournament level
- Round
- Best-of format

Weather Variables

- Temperature
- Humidity
- Wind speed
- Precipitation

Methods:

Prediction Setup

To create a consistent binary classification problem:

- **Player A** was defined as the **higher-ranked player entering the match**
- **Player B** was the lower-ranked player

Target variable:

- 1 = Player A wins
- 0 = Player A loses

Training and Testing Design

The dataset was randomly divided into:

- **80% Training Data**
- **20% Testing Data**

The split used stratification on the outcome variable, preserving the proportion of Player A wins and losses in both samples. This ensured that the training and testing sets had similar class balance and allowed fair model evaluation. All model tuning was performed using cross-validation on the training data only. The test set was reserved for final out-of-sample evaluation.

Models Estimated

1. Naive Benchmark

Always predict that Player A (the higher-ranked player) wins.

This provides a simple baseline showing how predictive rankings already are.

2. LASSO Logistic Regression

A penalized logistic regression model was estimated using LASSO regularization.

3. Gradient Boosted Trees (XGBoost)

A nonlinear model was estimated using gradient boosted decision trees. This model has the following advantages

- Captures nonlinear relationships
- Learns interactions automatically

Models were tuned around Accuracy, AUC, and mean log loss. The best model was selected by the model with the highest AUC.

Results:

Table 1 presents the out-of-sample performance of the naive benchmark, LASSO logistic regression, and gradient boosted tree (XGBoost) models.

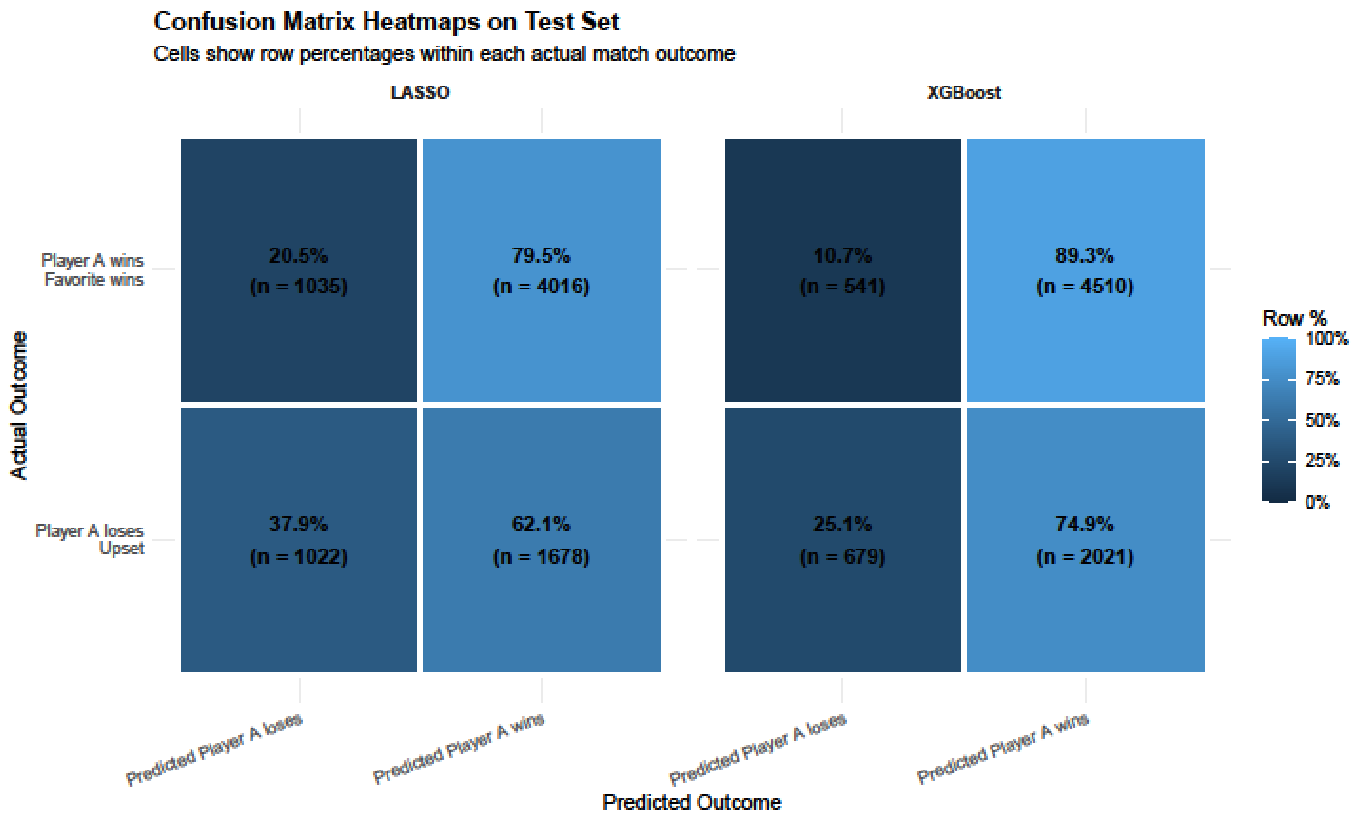
Table1

model	**accuracy**	**sensitivity**	**specificity**	**Roc auc**	**Mean log loss**
Naive: Always Pick Player A	0.651	1	0	0.5	12.556
LASSO	0.650	0.795	0.379	0.612	10.639
XGBoost	0.670	0.892	0.251	0.679	0.598

A simple benchmark model that always predicts the higher-ranked player (Player A) wins achieves an accuracy of approximately 65.1%. This result indicates that pre-match rankings alone provide substantial predictive power in ATP tennis matches, as higher-ranked players win the majority of contests.

The LASSO model does not meaningfully improve upon this baseline, achieving an accuracy of 65.0%. The gradient boosted tree model delivers the strongest overall predictive performance. XGBoost increases accuracy to 67.0%, representing a modest improvement over both the naive benchmark and the LASSO model. This improvement is more pronounced in the ROC AUC metric, where XGBoost achieves a value of 0.679 compared to 0.612 for LASSO and 0.500 for the naive benchmark. This suggests that the boosted tree model more effectively ranks match outcomes by probability.

Figure 1.



The models also exhibit clear differences in their prediction behavior. XGBoost achieves a high sensitivity of 0.892, indicating strong performance in correctly predicting matches where the higher-ranked player wins. However, its specificity remains relatively low at 0.251, suggesting limited ability to identify upset outcomes. Conversely, LASSO demonstrates a higher specificity but lower sensitivity, reflecting a greater willingness to predict upsets.

This can be visualized in the confusion matrix shown in figure 2.

Finally, mean log loss highlights substantial differences. The XGBoost model achieves a dramatically lower log loss (0.598) compared to LASSO (10.639) and the naive benchmark (12.556). This indicates that the boosted tree model produces more accurate and well-calibrated predicted probabilities, while LASSO tends to generate overly extreme and poorly calibrated predictions. I attempt to visualize this in figure 2.

Figure 2.

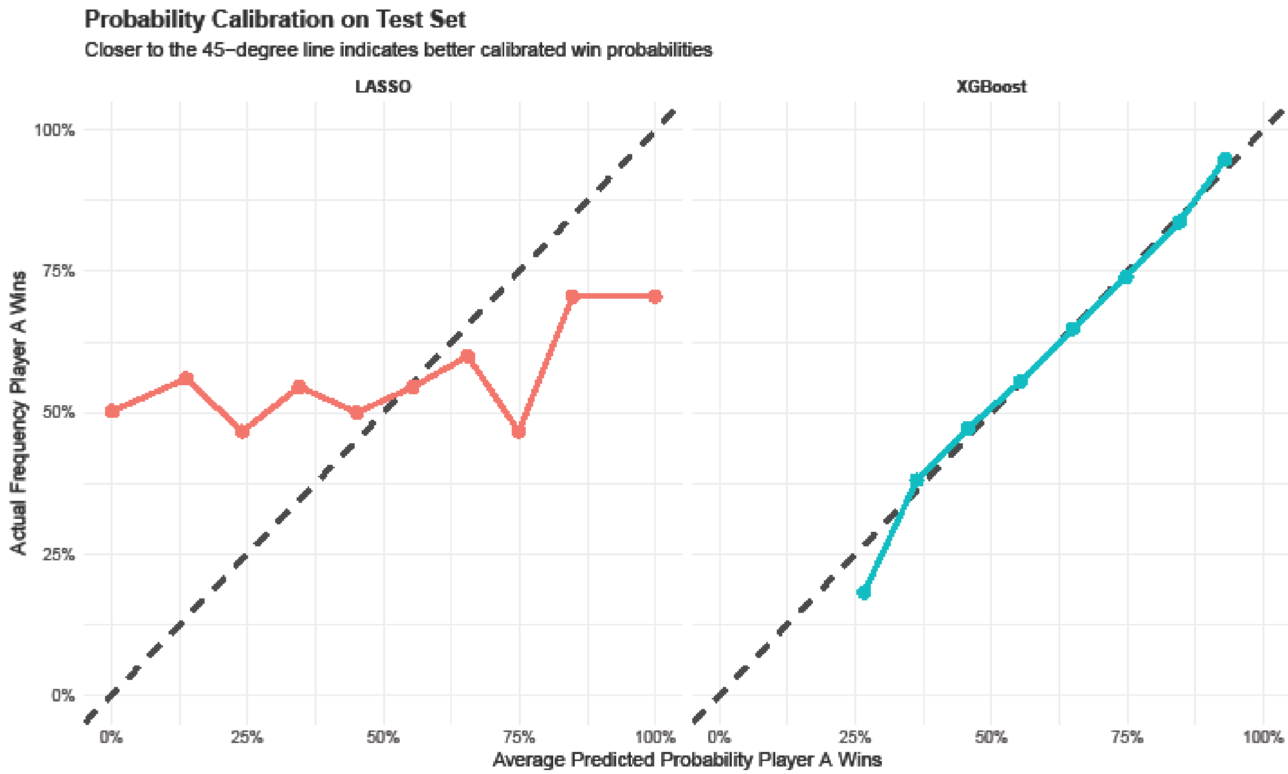


Figure 2 evaluates the calibration of predicted win probabilities for the LASSO and gradient boosted tree models. A perfectly calibrated model would produce predictions that lie along the 45-degree line, where predicted probabilities match observed win frequencies.

The LASSO model exhibits substantial deviations from this benchmark. Predicted probabilities are frequently extreme, yet the observed outcomes remain concentrated within a narrower range. For example, when the LASSO model assigns a near-zero probability that Player A wins, the actual win rate remains close to 50%. Similarly, when the model predicts near-certain victories, the observed win rate remains below 75%. These patterns indicate that the LASSO model is overconfident in its predictions.

In contrast, the gradient boosted tree model closely follows the 45-degree line across the range of predicted probabilities, indicating strong calibration. Predicted probabilities from the XGBoost model closely correspond to realized outcomes, suggesting that the model effectively captures uncertainty in match results. Notably, the model avoids assigning extreme probabilities near 0% or 100%, reflecting a more realistic assessment of outcome uncertainty.

These differences in calibration help explain the large gap in mean log loss reported in Table 1. Because log loss heavily penalizes confident but incorrect predictions.

While the xgboosts main improvement from the naïve baseline model is an improvement in the calibration of predicted probabilities it also does improve overall prediction accuracy. Figure 3 provides a visualization of which matchups the xgboost model outperforms the lasso and naïve baseline.

Figure 3.

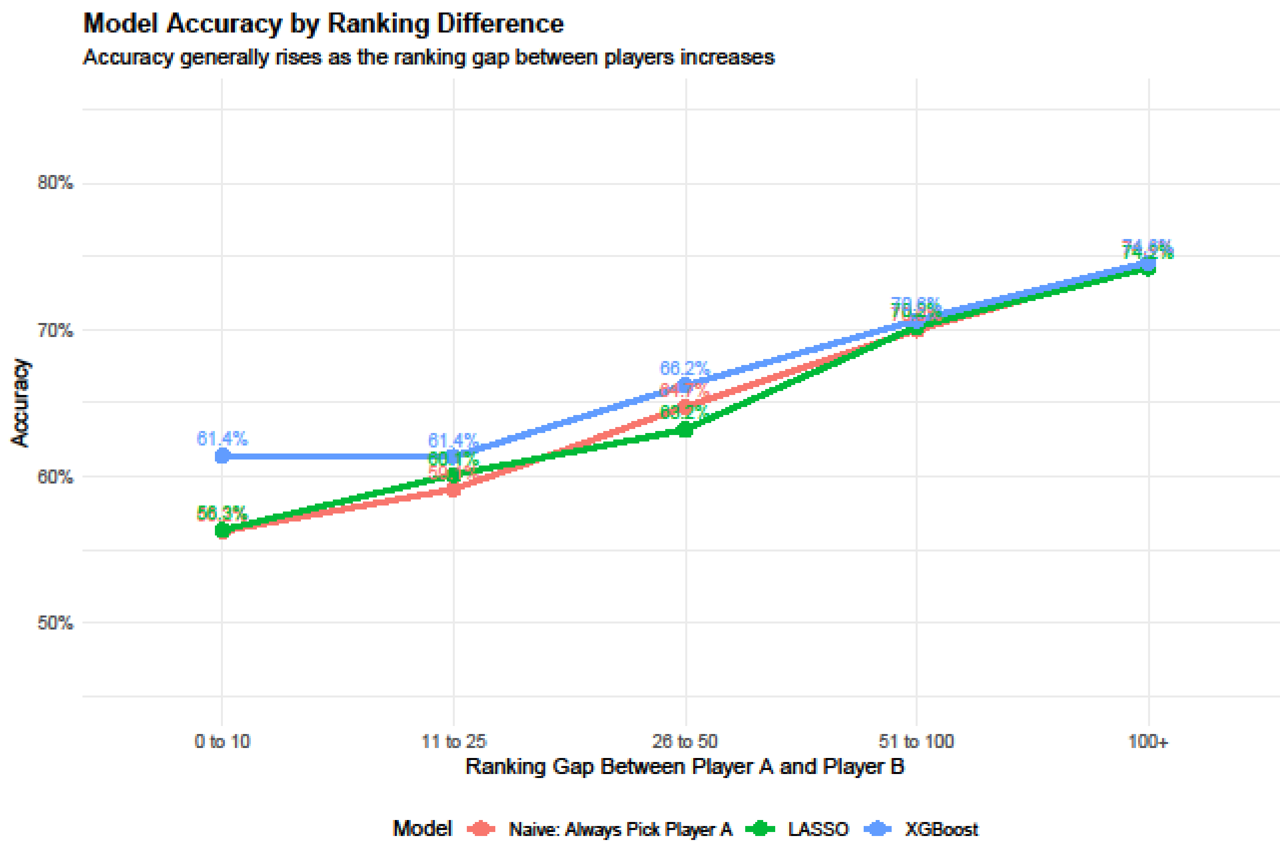
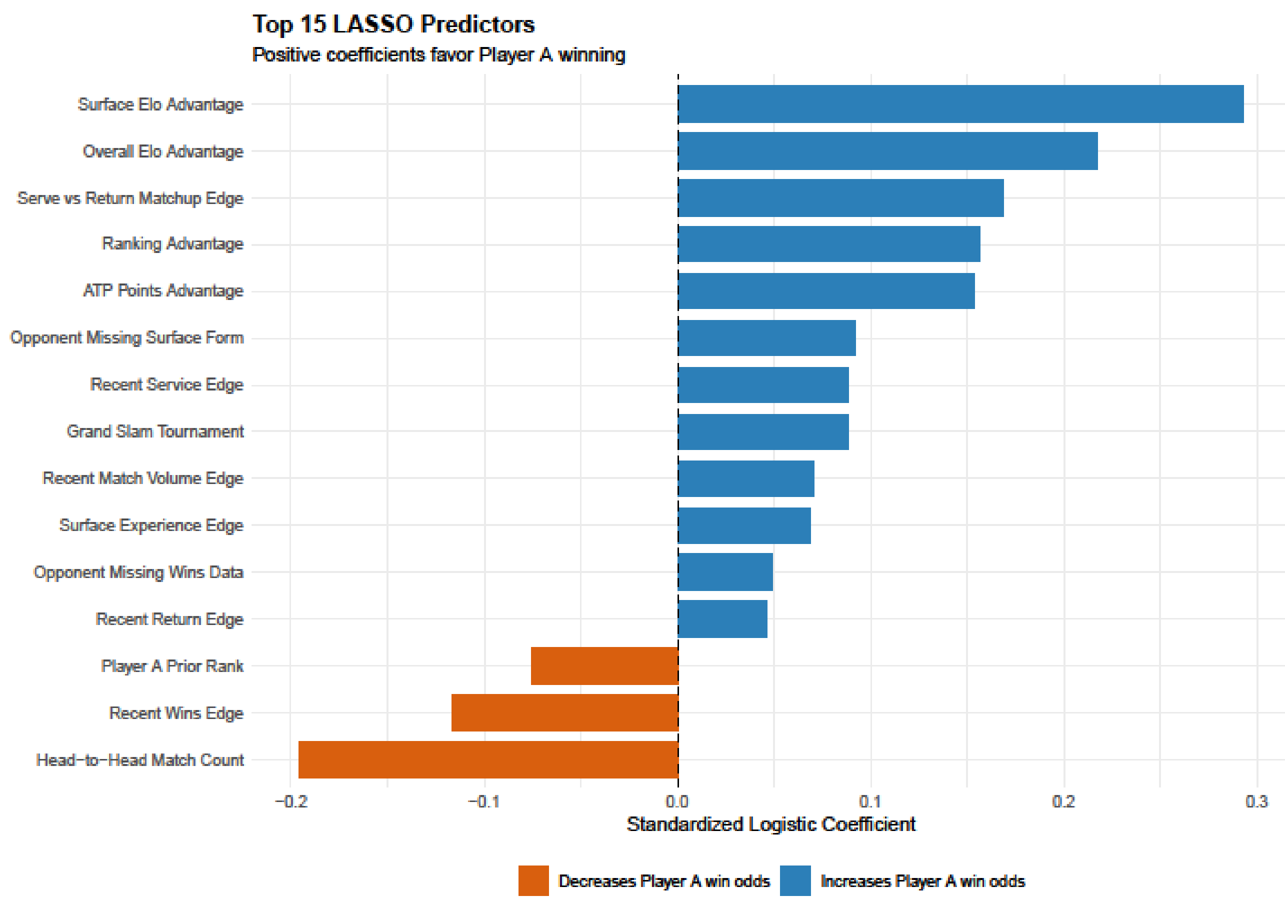
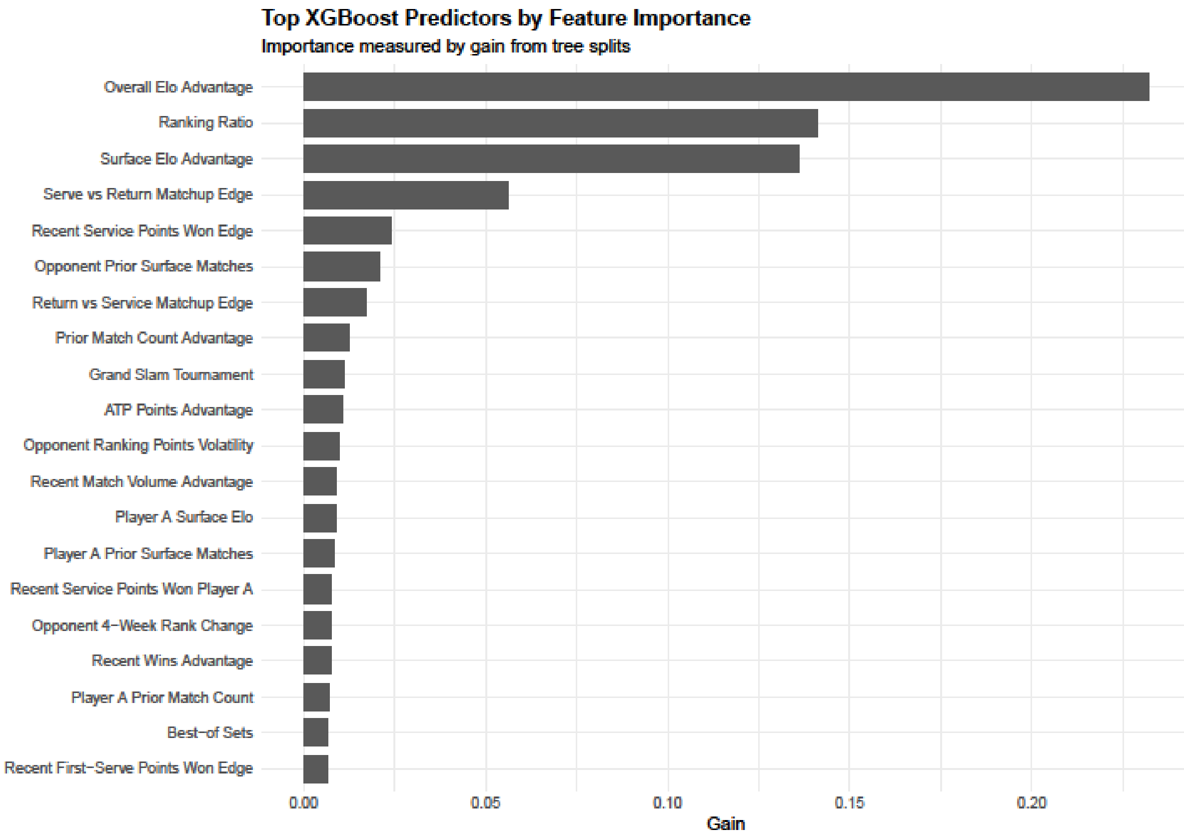


Figure 3 shows that the majority rank differences explain the majority of the predictive nature of tennis matches. As ranking differences grow the naïve model that simply always picks the higher ranked player to win converges with the xgboost model in accuracy. It is only when ranking differences are low, specifically between 0-10 that the xgboost model captures predictive information beyond ranks.

To better understand our models, figures 4 and 5 plot the most important predictors in both the lasso and xgboost models.


Top 15 LASSO Predictors
Positive coefficients favor Player A winning
Surface Elo Advantage
Overall Elo Advantage
Serve vs Return Matchup Edge
Ranking Advantage
ATP Points Advantage
Opponent Missing Surface Form
Recent Service Edge
Grand Slam Tournament
Recent Match Volume Edge
Surface Experience Edge
Opponent Missing Wins Data
Recent Return Edge
Player A Prior Rank
Recent Wins Edge
Head−to−Head Match Count
−0.2
−0.1
0.0
0.1
0.2
0.3
Standardized Logistic Coefficient
Decreases Player A win odds
Increases Player A win odds



Overall, these results suggest that rankings capture a large portion of the variation in match outcomes, nonlinear machine learning methods such as gradient boosted trees can extract additional predictive value. However, even the best-performing model struggles to accurately predict upset outcomes, highlighting the inherent uncertainty in professional tennis matches.